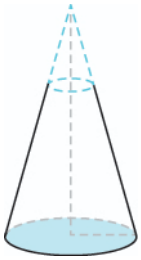
**302**. **Sand pile** What is the volume of a cone-shaped pile of sand that is $12$ meters tall and $30$ meters across at the base?

### Everyday Math

**303**. **Street light post** The post of a street light is shaped like a truncated cone, as shown in the picture below. It is a large cone minus a smaller top cone. The large cone is $30$ feet tall with base radius $1$ foot. The smaller cone is $10$ feet tall with base radius of $0.5$ feet. To the nearest tenth,
ⓐ find the volume of the large cone.

ⓑ find the volume of the small cone.

ⓒ find the volume of the post by subtracting the volume of the small cone from the volume of the large cone.



**304**. **Ice cream cones** A regular ice cream cone is $4$ inches tall and has a diameter of $2.5$ inches. A waffle cone is $7$ inches tall and has a diameter of $3.25$ inches. To the nearest hundredth,
ⓐ find the volume of the regular ice cream cone.

ⓑ find the volume of the waffle cone.

ⓒ how much more ice cream fits in the waffle cone compared to the regular cone?

### Writing Exercises

**305**. The formulas for the volume of a cylinder and a cone are similar. Explain how you can remember which formula goes with which shape.

**306**. Which has a larger volume, a cube of sides of $8$ feet or a sphere with a diameter of $8$ feet? Explain your reasoning.

### Self Check

*ⓐ After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

| I can... | Confidently | With some help | No-I don't get it! |
|---|---|---|---|
| find volume and surface area of rectangular solids. | | | |
| find volume and surface area of spheres. | | | |
| find volume and surface area of cylinders. | | | |
| find volume of cones. | | | |

*ⓑ After reviewing this checklist, what will you do to become confident for all objectives?*

## 9.7 Solve a Formula for a Specific Variable

### Learning Objectives

By the end of this section, you will be able to:

- Use the distance, rate, and time formula
- Solve a formula for a specific variable

☑ **BE PREPARED** 9.19 Before you get started, take this readiness quiz.

Write 35 miles per gallon as a unit rate.
If you missed this problem, review Example 5.65.

☑ **BE PREPARED** 9.20 Solve $6x + 24 = 96$.
If you missed this problem, review Example 8.20.

☑ **BE PREPARED** 9.21 Find the simple interest earned after 5 years on $\$1{,}000$ at an interest rate of $4\%$.
If you missed this problem, review Example 6.33.

## Use the Distance, Rate, and Time Formula

One formula you'll use often in algebra and in everyday life is the formula for distance traveled by an object moving at a constant speed. The basic idea is probably already familiar to you. Do you know what distance you travel if you drove at a steady rate of 60 miles per hour for 2 hours? (This might happen if you use your car's cruise control while driving on the Interstate.) If you said 120 miles, you already know how to use this formula!

The math to calculate the distance might look like this:

$$\text{distance} = \left(\frac{60 \text{ miles}}{1 \text{ hour}}\right)(2 \text{ hours})$$
$$\text{distance} = 120 \text{ miles}$$

In general, the formula relating distance, rate, and time is

$$\text{distance} = \text{rate} \cdot \text{time}$$

**Distance, Rate and Time**

For an object moving in at a uniform (constant) rate, the distance traveled, the elapsed time, and the rate are related by the formula

$$d = rt$$

where $d$ = distance, $r$ = rate, and $t$ = time.

Notice that the units we used above for the rate were miles per hour, which we can write as a ratio $\frac{miles}{hour}$. Then when we multiplied by the time, in hours, the common units 'hour' divided out. The answer was in miles.

**EXAMPLE 9.57**

Jamal rides his bike at a uniform rate of 12 miles per hour for $3\frac{1}{2}$ hours. How much distance has he traveled?

**Solution**

| | |
|---|---|
| Step 1. **Read** the problem.<br>You may want to create a mini-chart to summarize the information in the problem. | $d = ?$<br>$r = 12$ mph<br>$t = 3\frac{1}{2}$ hours |
| Step 2. **Identify** what you are looking for. | distance traveled |
| Step 3. **Name.** Choose a variable to represent it. | let $d$ = distance |
| Step 4. **Translate.**<br>Write the appropriate formula for the situation.<br>Substitute in the given information. | $d = rt$<br>$d = 12 \cdot 3\frac{1}{2}$ |

| | |
|---|---|
| Step 5. **Solve** the equation. | $d = 42$ miles |
| Step 6. **Check:** Does 42 miles make sense?<br>Jamal rides<br>12 miles in 1 hour,<br>24 miles in 2 hours,<br>36 miles in 3 hours,<br>48 miles in 4 hours,<br>42 miles in $3\frac{1}{2}$ hours is reasonable | |
| Step 7. **Answer** the question with a complete sentence. | Jamal rode 42 miles. |

**TRY IT** 9.113 Lindsay drove for $5\frac{1}{2}$ hours at 60 miles per hour. How much distance did she travel?

**TRY IT** 9.114 Trinh walked for $2\frac{1}{3}$ hours at 3 miles per hour. How far did she walk?

**EXAMPLE 9.58**

Rey is planning to drive from his house in San Diego to visit his grandmother in Sacramento, a distance of 520 miles. If he can drive at a steady rate of 65 miles per hour, how many hours will the trip take?

**Solution**

| | |
|---|---|
| Step 1. **Read** the problem.<br>Summarize the information in the problem. | $d = 520$ miles<br>$r = 65$ mph<br>$t = ?$ |
| Step 2. **Identify** what you are looking for. | how many hours (time) |
| Step 3. **Name:**<br>Choose a variable to represent it. | let $t$ = time |
| Step 4. **Translate.**<br>Write the appropriate formula.<br>Substitute in the given information. | $d = rt$<br>$520 = 65t$ |
| Step 5. **Solve** the equation. | $t = 8$ |
| Step 6. **Check:**<br>Substitute the numbers into the formula and make sure the result is a true statement.<br>$d = rt$<br>$520 \stackrel{?}{=} 65 \cdot 8$<br>$520 = 520$ ✓ | |
| Step 7. **Answer** the question with a complete sentence.<br>We know the units of time will be hours because we divided miles by miles per hour. | Rey's trip will take 8 hours. |

**TRY IT** 9.115 Lee wants to drive from Phoenix to his brother's apartment in San Francisco, a distance of 770 miles. If he drives at a steady rate of 70 miles per hour, how many hours will the trip take?

**TRY IT** 9.116 Yesenia is 168 miles from Chicago. If she needs to be in Chicago in 3 hours, at what rate does she need to drive?

## Solve a Formula for a Specific Variable

In this chapter, you became familiar with some formulas used in geometry. Formulas are also very useful in the sciences and social sciences—fields such as chemistry, physics, biology, psychology, sociology, and criminal justice. Healthcare workers use formulas, too, even for something as routine as dispensing medicine. The widely used spreadsheet program Microsoft Excel™ relies on formulas to do its calculations. Many teachers use spreadsheets to apply formulas to compute student grades. It is important to be familiar with formulas and be able to manipulate them easily.

In Example 9.57 and Example 9.58, we used the formula $d = rt$. This formula gives the value of $d$ when you substitute in the values of $r$ and $t$. But in Example 9.58, we had to find the value of $t$. We substituted in values of $d$ and $r$ and then used algebra to solve to $t$. If you had to do this often, you might wonder why there isn't a formula that gives the value of $t$ when you substitute in the values of $d$ and $r$. We can get a formula like this by solving the formula $d = rt$ for $t$.

**To solve a formula for a specific variable** means to get that variable by itself with a coefficient of 1 on one side of the equation and all the other variables and constants on the other side. We will call this solving an equation for a specific variable *in general.* This process is also called *solving a literal equation*. The result is another formula, made up only of variables. The formula contains letters, or *literals*.

Let's try a few examples, starting with the distance, rate, and time formula we used above.

**EXAMPLE 9.59**

Solve the formula $d = rt$ for $t$:

ⓐ when $d = 520$ and $r = 65$ ⓑ in general.

**Solution**

We'll write the solutions side-by-side so you can see that solving a formula in general uses the same steps as when we have numbers to substitute.

| | ⓐ when $d = 520$ and $r = 65$ | ⓑ in general |
|---|---|---|
| Write the forumla. | $d = rt$ | $d = rt$ |
| Substitute any given values. | $520 = 65t$ | |
| Divide to isolate $t$. | $\frac{520}{65} = \frac{65t}{65}$ | $\frac{d}{r} = \frac{rt}{r}$ |
| Simplify. | $8 = t$<br>$t = 8$ | $\frac{d}{r} = t$<br>$t = \frac{d}{r}$ |

Notice that the solution for ⓐ is the same as that in Example 9.58. We say the formula $t = \frac{d}{r}$ is solved for $t$. We can use this version of the formula anytime we are given the distance and rate and need to find the time.

**TRY IT** 9.117 Solve the formula $d = rt$ for $r$:

ⓐ when $d = 180$ and $t = 4$ ⓐ in general

**TRY IT** 9.118 Solve the formula $d = rt$ for $r$:

ⓐ when $d = 780$ and $t = 12$ ⓑ in general

We used the formula $A = \frac{1}{2}bh$ in Use Properties of Rectangles, Triangles, and Trapezoids to find the area of a triangle when we were given the base and height. In the next example, we will solve this formula for the height.

**EXAMPLE 9.60**

The formula for area of a triangle is $A = \frac{1}{2}bh$. Solve this formula for $h$:

ⓐ when $A = 90$ and $b = 15$ ⓑ in general

**Solution**

| | ⓐ when $A = 90$ and $b = 15$ | ⓑ in general |
|---|---|---|
| Write the forumla. | $A = \frac{1}{2}bh$ | $A = \frac{1}{2}bh$ |
| Substitute any given values. | $90 = \frac{1}{2} \cdot 15 \cdot h$ | |
| Clear the fractions. | $2 \cdot 90 = 2 \cdot \frac{1}{2} \cdot 15 \cdot h$ | $2 \cdot A = 2 \cdot \frac{1}{2} \cdot b \cdot h$ |
| Simplify. | $180 = 15h$ | $2A = bh$ |
| Solve for $h$. | $12 = h$ | $\frac{2A}{b} = h$ |

We can now find the height of a triangle, if we know the area and the base, by using the formula

$$h = \frac{2A}{b}$$

**TRY IT** 9.119 Use the formula $A = \frac{1}{2}bh$ to solve for h:

ⓐ when $A = 170$ and $b = 17$ ⓑ in general

**TRY IT** 9.120 Use the formula $A = \frac{1}{2}bh$ to solve for $b$:

ⓐ when $A = 62$ and $h = 31$ ⓑ in general

In Solve Simple Interest Applications, we used the formula $I = Prt$ to calculate simple interest, where $I$ is interest, $P$ is principal, $r$ is rate as a decimal, and $t$ is time in years.

**EXAMPLE 9.61**

Solve the formula $I = Prt$ to find the principal, $P$:

ⓐ when $I = \$5{,}600$, $r = 4\%$, $t = 7$ years ⓑ in general

**Solution**

| | $I$ = \$5600, $r$ = 4%, $t$ = 7 years | in general |
|---|---|---|
| Write the forumla. | $I = Prt$ | $I = Prt$ |
| Substitute any given values. | $5600 = P(0.04)(7)$ | $I = Prt$ |
| Multiply $r \cdot t$. | $5600 = P(0.28)$ | $I = P(rt)$ |
| Divide to isolate $P$. | $\frac{5600}{0.28} = \frac{P(0.28)}{0.28}$ | $\frac{I}{rt} = \frac{P(rt)}{rt}$ |
| Simplify. | $20{,}000 = P$ | $\frac{I}{rt} = P$ |
| State the answer. | The principal is \$20,000. | $P = \frac{I}{rt}$ |

**TRY IT** 9.121 Use the formula $I = Prt$.

Find $t$: (a) when $I = \$2{,}160,\ r = 6\%,\ P = \$12{,}000$; (b) in general

**TRY IT** 9.122 Use the formula $I = Prt$.

Find $r$: (a) when $I = \$5{,}400,\ P = \$9{,}000,\ t = 5$ years (b) in general

Later in this class, and in future algebra classes, you'll encounter equations that relate two variables, usually $x$ and $y$. You might be given an equation that is solved for $y$ and need to solve it for $x$, or vice versa. In the following example, we're given an equation with both $x$ and $y$ on the same side and we'll solve it for $y$. To do this, we will follow the same steps that we used to solve a formula for a specific variable.

**EXAMPLE 9.62**

Solve the formula $3x + 2y = 18$ for $y$:

(a) when $x = 4$ (b) in general

**Solution**

| | when $x$ = 4 | in general |
|---|---|---|
| Write the equation. | $3x + 2y = 18$ | $3x + 2y = 18$ |
| Substitute any given values. | $3(4) + 2y = 18$ | $3x + 2y = 18$ |
| Simplify if possible. | $12 + 2y = 18$ | $3x + 2y = 18$ |
| Subtract to isolate the $y$-term. | $12 - 12 + 2y = 18 - 12$ | $3x - 3x + 2y = 18 - 3x$ |

| Simplify. | $2y = 6$ | $2y = 18 - 3x$ |
|---|---|---|
| Divide. | $\frac{2y}{2} = \frac{6}{2}$ | $\frac{2y}{2} = \frac{18 - 3x}{2}$ |
| Simplify. | $y = 3$ | $y = \frac{18 - 3x}{2}$ |

**TRY IT** 9.123 Solve the formula $3x + 4y = 10$ for $y$:

ⓐ when $x = 2$ ⓑ in general

**TRY IT** 9.124 Solve the formula $5x + 2y = 18$ for $y$:

ⓐ when $x = 4$ ⓑ in general

In the previous examples, we used the numbers in part (a) as a guide to solving in general in part (b). Do you think you're ready to solve a formula in general without using numbers as a guide?

**EXAMPLE 9.63**

Solve the formula $P = a + b + c$ for $a$.

**Solution**

We will isolate $a$ on one side of the equation.

| We will isolate *a* on one side of the equation. | |
|---|---|
| Write the equation. | $P = a + b + c$ |
| Subtract *b* and *c* from both sides to isolate *a*. | $P - b - c = a + b + c - b - c$ |
| Simplify. | $P - b - c = a$ |

So, $a = P - b - c$

**TRY IT** 9.125 Solve the formula $P = a + b + c$ for $b$.

**TRY IT** 9.126 Solve the formula $P = a + b + c$ for $c$.

**EXAMPLE 9.64**

Solve the equation $3x + y = 10$ for $y$.

**Solution**

We will isolate $y$ on one side of the equation.

| We will isolate $y$ on one side of the equation. | |
|---|---|
| Write the equation. | $3x + y = 10$ |
| Subtract $3x$ from both sides to isolate $y$. | $3x - 3x + y = 10 - 3x$ |
| Simplify. | $y = 10 - 3x$ |

**TRY IT** 9.127 Solve the formula $7x + y = 11$ for $y$.

**TRY IT** 9.128 Solve the formula $11x + y = 8$ for $y$.

**EXAMPLE 9.65**

Solve the equation $6x + 5y = 13$ for $y$.

**Solution**

We will isolate $y$ on one side of the equation.

| We will isolate $y$ on one side of the equation. | |
|---|---|
| Write the equation. | $6x + 5y = 13$ |
| Subtract to isolate the term with $y$. | $6x + 5y - 6x = 13 - 6x$ |
| Simplify. | $5y = 13 - 6x$ |
| Divide 5 to make the coefficient 1. | $\frac{5y}{5} = \frac{13 - 6x}{5}$ |
| Simplify. | $y = \frac{13 - 6x}{5}$ |

**TRY IT** 9.129 Solve the formula $4x + 7y = 9$ for $y$.

**TRY IT** 9.130 Solve the formula $5x + 8y = 1$ for $y$.

**LINKS TO LITERACY**

The Links to Literacy activity *What's Faster than a Speeding Cheetah?* will provide you with another view of the topics covered in this section.

**MEDIA**

**ACCESS ADDITIONAL ONLINE RESOURCES**

Distance = Rate x Time (https://www.openstax.org/l/24distratextime)

Distance, Rate, Time (https://www.openstax.org/l/24distratetime)

Simple Interest (https://www.openstax.org/l/24simpinterest)
Solving a Formula for a Specific Variable (https://www.openstax.org/l/24solvespvari)
Solving a Formula for a Specific Variable (https://www.openstax.org/l/24solvespecivar)

## SECTION 9.7 EXERCISES

### Practice Makes Perfect

#### *Use the Distance, Rate, and Time Formula*

*In the following exercises, solve.*

**307.** Steve drove for $8\frac{1}{2}$ hours at $72$ miles per hour. How much distance did he travel?

**308.** Socorro drove for $4\frac{5}{6}$ hours at $60$ miles per hour. How much distance did she travel?

**309.** Yuki walked for $1\frac{3}{4}$ hours at $4$ miles per hour. How far did she walk?

**310.** Francie rode her bike for $2\frac{1}{2}$ hours at $12$ miles per hour. How far did she ride?

**311.** Connor wants to drive from Tucson to the Grand Canyon, a distance of $338$ miles. If he drives at a steady rate of $52$ miles per hour, how many hours will the trip take?

**312.** Megan is taking the bus from New York City to Montreal. The distance is $384$ miles and the bus travels at a steady rate of $64$ miles per hour. How long will the bus ride be?

**313.** Aurelia is driving from Miami to Orlando at a rate of $65$ miles per hour. The distance is $235$ miles. To the nearest tenth of an hour, how long will the trip take?

**314.** Kareem wants to ride his bike from St. Louis, Missouri to Champaign, Illinois. The distance is $180$ miles. If he rides at a steady rate of $16$ miles per hour, how many hours will the trip take?

**315.** Javier is driving to Bangor, Maine, which is $240$ miles away from his current location. If he needs to be in Bangor in $4$ hours, at what rate does he need to drive?

**316.** Alejandra is driving to Cincinnati, Ohio, $450$ miles away. If she wants to be there in $6$ hours, at what rate does she need to drive?

**317.** Aisha took the train from Spokane to Seattle. The distance is $280$ miles, and the trip took $3.5$ hours. What was the speed of the train?

**318.** Philip got a ride with a friend from Denver to Las Vegas, a distance of $750$ miles. If the trip took $10$ hours, how fast was the friend driving?

#### *Solve a Formula for a Specific Variable*

*In the following exercises, use the formula. $d = rt$.*

**319.** Solve for $t$:
(a) when $d = 350$ and $r = 70$
(b) in general

**320.** Solve for $t$:
(a) when $d = 240$ and $r = 60$
(b) in general

**321.** Solve for $t$:
(a) when $d = 510$ and $r = 60$
(b) in general

**322.** Solve for $t$:
(a) when $d = 175$ and $r = 50$
(b) in general

**323.** Solve for $r$:
(a) when $d = 204$ and $t = 3$
(b) in general

**324.** Solve for $r$:
(a) when $d = 420$ and $t = 6$
(b) in general

**325.** Solve for $r$:
(a) when $d = 160$ and $t = 2.5$
(b) in general

**326.** Solve for $r$:
(a) when $d = 180$ and $t = 4.5$
(b) in general.

*In the following exercises, use the formula $A = \frac{1}{2}bh$.*

**327.** Solve for $b$:
(a) when $A = 126$ and $h = 18$
(b) in general

**328.** Solve for $h$:
(a) when $A = 176$ and $b = 22$
(b) in general

**329.** Solve for $h$:
(a) when $A = 375$ and $b = 25$
(b) in general

**330.** Solve for $b$:
(a) when $A = 65$ and $h = 13$
(b) in general

*In the following exercises, use the formula $I = Prt$.*

**331.** Solve for the principal, $P$ for:
(a) $I = \$5{,}480$, $r = 4\%$, $t = 7$ years
(b) in general

**332.** Solve for the principal, $P$ for:
(a) $I = \$3{,}950$, $r = 6\%$, $t = 5$ years
(b) in general

**333.** Solve for the time, $t$ for:
(a) $I = \$2{,}376$, $P = \$9{,}000$, $r = 4.4\%$
(b) in general

**334.** Solve for the time, $t$ for:
(a) $I = \$624$, $P = \$6{,}000$, $r = 5.2\%$
(b) in general

*In the following exercises, solve.*

**335.** Solve the formula $2x + 3y = 12$ for $y$:
(a) when $x = 3$
(b) in general

**336.** Solve the formula $5x + 2y = 10$ for $y$:
(a) when $x = 4$
(b) in general

**337.** Solve the formula $3x + y = 7$ for $y$:
(a) when $x = -2$
(b) in general

**338.** Solve the formula $4x + y = 5$ for $y$:
(a) when $x = -3$
(b) in general

**339.** Solve $a + b = 90$ for $b$.

**340.** Solve $a + b = 90$ for $a$.

**341.** Solve $180 = a + b + c$ for $a$.

**342.** Solve $180 = a + b + c$ for $c$.

**343.** Solve the formula $8x + y = 15$ for $y$.

**344.** Solve the formula $9x + y = 13$ for $y$.

**345.** Solve the formula $-4x + y = -6$ for $y$.

**346.** Solve the formula $-5x + y = -1$ for $y$.

**347.** Solve the formula $4x + 3y = 7$ for $y$.

**348.** Solve the formula $3x + 2y = 11$ for $y$.

**349.** Solve the formula $x - y = -4$ for $y$.

**350.** Solve the formula $x - y = -3$ for $y$.

**351.** Solve the formula $P = 2L + 2W$ for $L$.

**352.** Solve the formula $P = 2L + 2W$ for $W$.

**353.** Solve the formula $C = \pi d$ for $d$.

**354.** Solve the formula $C = \pi d$ for $\pi$.

**355.** Solve the formula $V = LWH$ for $L$.

**356.** Solve the formula $V = LWH$ for $H$.

## Everyday Math

**357**. **Converting temperature** While on a tour in Greece, Tatyana saw that the temperature was $40°$ Celsius. Solve for $F$ in the formula $C = \frac{5}{9}(F - 32)$ to find the temperature in Fahrenheit.

**358**. **Converting temperature** Yon was visiting the United States and he saw that the temperature in Seattle was $50°$ Fahrenheit. Solve for $C$ in the formula $F = \frac{9}{5}C + 32$ to find the temperature in Celsius.

## Writing Exercises

**359**. Solve the equation $2x + 3y = 6$ for $y$:
ⓐ when $x = -3$ ⓑ in general
ⓒ Which solution is easier for you? Explain why.

**360**. Solve the equation $5x - 2y = 10$ for $x$:
ⓐ when $y = 10$ ⓑ in general
ⓒ Which solution is easier for you? Explain why.

## Self Check

ⓓ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

| I can... | Confidently | With some help | No-I don't get it! |
|---|---|---|---|
| use the distance, rate, and time formula. | | | |
| solve a formula for a specific variable. | | | |

ⓑ *Overall, after looking at the checklist, do you think you are well-prepared for the next Chapter? Why or why not?*